**Sucrase**: Sucrase which acts upon sucrose to cleave it into equimolar quantities of glucose and fructose has an optimum pH of 6.

**Isomaltase:** Isomaltase co-elutes with Sucrase on purification and therefore Sucrase/ isomaltase is a complex protein. Isomaltase acts on the α1→6 linkage and cleaves it into two molecules of glucose.

$$\textit{Isomall}\text{tose} \xrightarrow{\text{Isomaltase}} \text{Glucose+ Glucose}$$

**Lactase:** The β 1→4 glycosidic linkage of lactose is acted upon by Lactase and cleaves it into equimolar quantities of glucose and galactose. The optimum pH for *sucrase* is the optimal pH for *lactase* which is around 6, but it can function in an acidic environment ranging between a pH of 2 to 7.

$$\text{lactose} \xrightarrow{\text{Lactase}} \text{Glucose + Galactose}$$

**Maltase:** Maltose, the end product of polysaccharide digestion by amylases is cleaved into two glucose molecules by cleavage of α1→4 linkage by maltase to liberate glucose into the lumen. It also has an optimum pH around 6 (Figure 2.13).

$$\textit{Mal}\text{tose} \xrightarrow{\text{Maltase}} \text{Glucose+ Glucose}$$

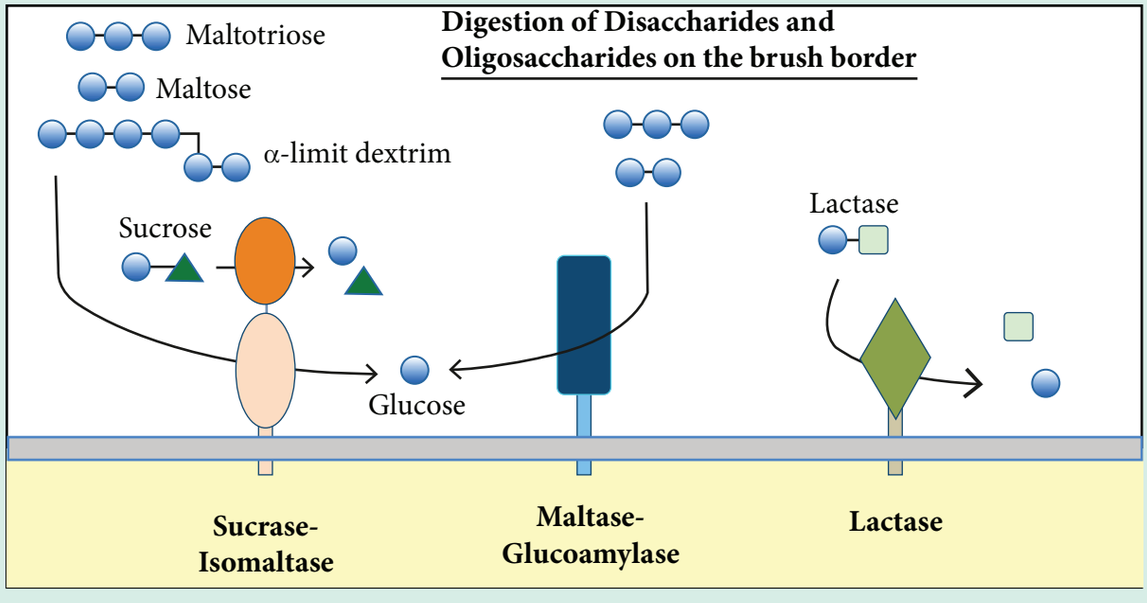



**Figure 2.13 Digestion of disaccharides on the brush border**

**Cerebrosidases :** Cerebrosidases are a group of minor enzymes that hydrolyse the gluco and galactocerebrosides present in glycolipids.



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